



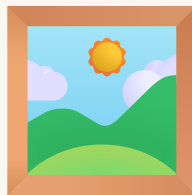
Designing the Museum

Plan the coordinates before you build

짓기 전에 좌표부터 설계해요 — 설계도

BIG PICTURE

One room, everything together



Floor (x, z) + wall art (x, y) + pedestals (x, y, z).

바닥 · 벽 그림 · 받침대를 한 방에

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a blueprint ?

A plan that lists **every coordinate** before you build — so nothing overlaps and each statue lands on its stand.

미리 좌표를 정해두면 겹치지 않아요

CONCEPT

Each piece, its own coordinates

floor
(x, z) flat

picture
(x, y) on wall

pedestal
(x, y, z) box

바닥은 (x,z), 그림은 (x,y), 받침대는 (x,y,z)

CODE

Pedestal + statue on top

```
# pedestal fills up to height 1  
blocks.fill(STONE, pos(2,0,2), pos(4,1,4), FillOperation.REPLACE)  
# statue rests one block higher  
blocks.place(GOLD_BLOCK, pos(3, 2, 3))
```

Stone ends at $y = 1$, so the statue sits at $y = 2$ — right on top.

CODE

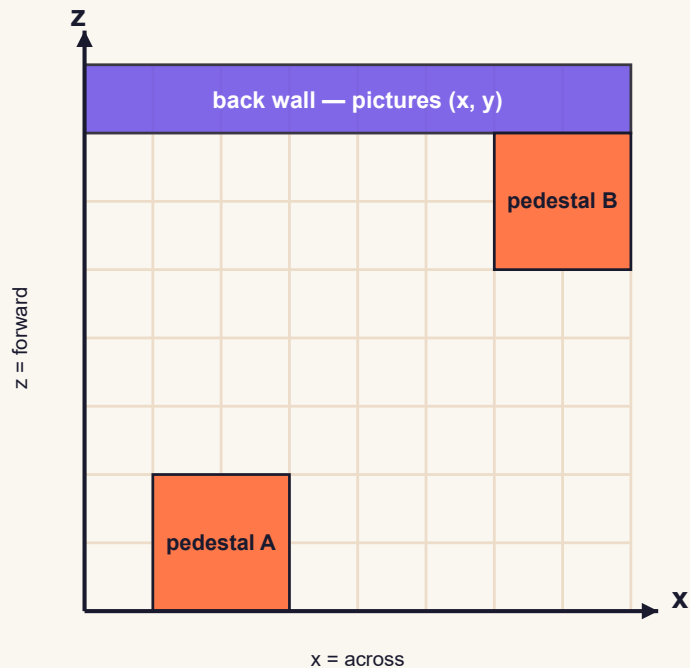
Two pedestals, no overlap

```
blocks.fill(STONE, pos(1,0,1), pos(2,1,2), FillOperation.REPLACE)  
blocks.fill(STONE, pos(6,0,6), pos(7,1,7), FillOperation.REPLACE)
```

Different **(x, z)** footprints — one near front-left, one far back-right. They never crash.

SEE IT

Blueprint — looking down from above



위에서 본 설계도 — 받침대 두 개의 (x, z) 자리가 겹치지 않게 미리 정해요

PREDICT 🤖

Does the statue rest on top?

Predict first — then click reveal

```
pedestal: fill stone from (2, 0, 2) to (4, 1, 4)
statue:   place gold block at (3, 2, 3)
```

Reveal output 🔍 reveal

VOCAB

New words today

blueprint a coordinate plan made before building

pedestal a short box a sculpture sits on

sculpture a 3D artwork on a pedestal

overlap two pieces sharing the same footprint

설계도

받침대

조각상

겹침



DEBUG

What's wrong with this code?

Goal: the statue **rests on top** of the pedestal (top is at $y = 1$).

buggy

```
fill stone from (2, 0, 2) to (4, 1, 4)
place gold block at (3, 5, 3)
```

Think first — then click next.



DEBUG

The bug

Bug: $y = 5$ floats the statue high in the air. The pedestal ends at height 1, so the top block is $y = 2$, not 5.

fixed

```
fill stone from (2, 0, 2) to (4, 1, 4)
place gold block at (3, 2, 3)
```

YOUR TURN!

Draw a blueprint

- ✓ Sketch one room from above — floor, back wall, pedestals
- ✓ Write the floor's two corner coordinates
- ✓ Give **two** pedestals different (x, z) footprints
- ✓ Write each statue's (x, y, z) resting on top
- ✓ Build it from your blueprint, then send a photo



QUIZ

QUESTION 1 OF 3

Why make a blueprint **before** building?

So the build runs faster.

It's not needed at all.

So coordinates are planned and nothing overlaps.

To pick block colours only.



QUESTION 2 OF 3

A pedestal fills up to height 1. Where does the statue rest?

$y = 2$

$y = 5$

$y = 1$

$y = 0$



QUIZ

QUESTION 3 OF 3

How do you stop two pedestals from crashing into each other?

Place them at the same coordinates.

Give them different (x, z) footprints.

Use gold instead of stone.

Make them the same height.

QUIZ

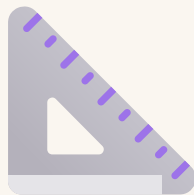
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Blueprint ready!

Next: build your own museum.

다음 주: 나만의 미술관 완성